THE MASTERMIND COACHING CLASSES

CBSE / RBSE · Nursery – Class XII · Road No. 5, VKIA, Jaipur · 9887258679

ANSWER KEY & MARKING SCHEME — MATHS · CLASS 9 · SET 04

Max. Marks: 80 · The Mastermind Coaching Classes, Jaipur · Alternate correct methods to be given full credit

SECTION A — Objective Type — 20 questions · 20 marks

Q1–Q18 are multiple-choice questions (1 mark each). Q19–Q20 are Assertion (A) and Reason (R) questions: select the correct option from (a), (b), (c), (d) given below each.

1. The abscissa of every point lying on the y-axis is:

(a) equal to its ordinate (b) 0

(c) 1 (d) undefined

Ans: (b) A point on the y-axis has the form (0, y).

2. If (x − 2, y + 5) = (0, 0), then the value of x + y is:

(a) −3 (b) 7 (c) −7 (d) 3

Ans: (a) x = 2, y = −5 ⇒ x + y = −3.

3. The midpoint of the segment joining (−6, 3) and (4, −9) is:

(a) (−1, −3) (b) (−5, −6) (c) (1, −3) (d) (−1, 3)

Ans: (a) ((−6 + 4)/2, (3 − 9)/2) = (−1, −3).

4. The value of the polynomial 2x − 7 at x = −3 is:

(a) 13 (b) −13 (c) −1 (d) −9

Ans: (b) 2(−3) − 7 = −13.

5. The zero of the polynomial 3x + 9 is:

(a) 3 (b) −9 (c) 9 (d) −3

Ans: (d) 3x + 9 = 0 ⇒ x = −3.

6. The side of an equilateral triangle is (2x − 1) cm. Its perimeter (in cm) is:

(a) 8x − 4 (b) 6x − 3 (c) 2x − 3 (d) 6x − 1

Ans: (b) 3(2x − 1) = 6x − 3.

7. Which of the following is a rational number?

(a) √(49/16) (b) √7

(c) π (d) 0.12112111211112…

Ans: (a) √(49/16) = 7/4 is rational; the others are irrational.

8. The decimal expansion of $\frac{7}{16}$ terminates after how many decimal places?

(a) 5 (b) 3 (c) 4 (d) 2

Ans: (c) 7/16 = 0.4375.

9. The value of $|-\frac{2}{3}| \times |\frac{9}{4}|$ is:

(a) $\frac{1}{6}$ (b) $\frac{8}{27}$ (c) $\frac{-3}{2}$ (d) $\frac{3}{2}$

Ans: (d) 2/3 × 9/4 = 3/2.

10. On expanding $(2x + 5)^2$, we get:

(a) $4x^2 + 25$ (b) $2x^2 + 20x + 25$ (c) $4x^2 + 20x + 25$ (d) $4x^2 + 10x + 25$

Ans: (c) $(a + b)^2 = a^2 + 2ab + b^2$.

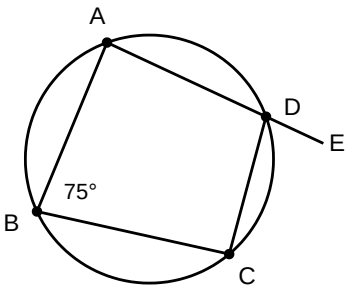
11. The factorisation of $1 - 49a^2$ is:

(a) $(1 - 49a)(1 + a)$ (b) $(1 - 7a)^2$

(c) $(1 - 7a)(1 + 7a)$ (d) $(7a - 1)^2$

Ans: (c) $1^2 - (7a)^2$.

12. In the figure, ABCD is a cyclic quadrilateral and the side AD is produced to E. If $\angle ABC = 75°$, then the exterior angle $\angle CDE$ is:



(a) 105° (b) 15° (c) 75° (d) 150°

Ans: (c) $\angle ADC = 180° - 75° = 105°$; $\angle CDE = 180° - 105° = 75°$ (exterior angle = interior opposite angle).

13. The longest chord of a circle of radius 5 cm has length:

(a) 2.5 cm (b) 20 cm (c) 5 cm (d) 10 cm

Ans: (d) The diameter, $2 \times 5 = 10$ cm.

14. The area of a right-angled triangle with sides 6 cm, 8 cm and 10 cm, found using Heron's formula, is:

(a) 24 cm² (b) 30 cm² (c) 48 cm² (d) 40 cm²

Ans: (a) $s = 12$; $\sqrt{(12 \times 6 \times 4 \times 2)} = \sqrt{576} = 24$.

15. The area of an isosceles triangle with sides 25 cm, 25 cm and 14 cm is:

(a) 84 cm² (b) 175 cm² (c) 336 cm² (d) 168 cm²

Ans: (d) $s = 32$; $\sqrt{(32 \times 7 \times 7 \times 18)} = \sqrt{28224} = 168$.

16. A number is picked at random from the numbers 1 to 25. The probability that it is a prime number is:

(a) $\frac{9}{25}$ (b) $\frac{8}{25}$ (c) $\frac{10}{25}$ (d) $\frac{12}{25}$

Ans: (a) Primes ≤ 25: 2, 3, 5, 7, 11, 13, 17, 19, 23 = 9 numbers.

17. A coin and a die are thrown together. The number of possible outcomes is:

(a) 8 (b) 6 (c) 36 (d) 12

Ans: (d) $2 \times 6 = 12$.

18. The sum of the first 10 natural numbers is:

(a) 110 (b) 55 (c) 45 (d) 50

Ans: (b) $10 \times 11 / 2 = 55$.

19. **Assertion (A):** The sequence 5, 8, 11, 14, … is an AP with common difference 3.
Reason (R): In an AP the difference between consecutive terms is constant.

Ans: (a) $8 - 5 = 11 - 8 = 14 - 11 = 3$.

20. **Assertion (A):** The area of a triangle with sides 7 cm, 8 cm and 20 cm can be found using Heron's formula.
Reason (R): The sum of any two sides of a triangle is greater than its third side.

> **Ans: (d)** 7 + 8 < 20, so no such triangle exists ⇒ A false; R is the true triangle-inequality statement.

SECTION B — Very Short Answer (2 marks each) — 5 questions · 10 marks

Internal choice is provided in one question.

21. A plumber charges ₹150 as visiting charge and ₹75 for every hour of work. Write the expression for the total charge for h hours and find it for 4 hours.

> Charge = 75h + 150 (1); for h = 4: 300 + 150 = **₹450** (1)

22. Without actual division, show that $\frac{51}{1600}$ has a terminating decimal expansion and state the number of decimal places.

> $1600 = 2^6 \times 5^2$ (only 2 and 5) ⇒ terminating (1). Decimal places = greater power = **6** (1).

O R

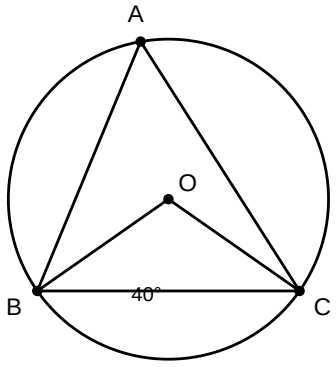
Find two rational numbers between 0.1 and 0.2.

> Any two, e.g. 0.12 and 0.15 (1 each); (check: 0.1 < 0.12 < 0.15 < 0.2).

23. Using a suitable identity, evaluate 1002 × 998.

> $1002 \times 998 = (1000 + 2)(1000 - 2)$ (1) = $1000^2 - 2^2$ = **999996** (1)

24. In the figure, O is the centre of the circle and ∠OBC = 40°. Find ∠BAC.



> OB = OC ⇒ ∠OCB = ∠OBC = 40°, so ∠BOC = 100° (1)
> ∠BAC = ½ ∠BOC = **50°** (1)

25. Find the second, third and fourth terms of the sequence defined by $t_1 = 7$ and $t_n = t_{n-1} - 3$ for $n \geq 2$.

> $t_2 = 4$, $t_3 = 1$, $t_4 = -2$. (1 for method, 1 for values)

SECTION C — Short Answer (3 marks each) — 6 questions · 18 marks

Internal choice is provided in two questions.

26. Show that $0.285714285714\ldots = \frac{2}{7}$.

> Let x = 0.285714285714…; $10^6 x = 285714.285714\ldots$ (1)
> 999999x = 285714 (1) ⇒ x = 285714/999999 = **2/7** (dividing by 142857) (1)

27. 200 students were asked which type of book they borrowed:

Type	Fiction	Science	History	Comics
Students	70	50	30	50

A student is chosen at random. Find the probability that the student borrowed (i) fiction, (ii) science or history, (iii) not comics.

(i) 70/200 = **7/20** (1) (ii) 80/200 = **2/5** (1) (iii) 150/200 = **3/4** (1)

OR

The number of goals scored by a team in 40 matches:

Goals	0	1	2	3
Matches	6	14	12	8

A match is chosen at random. Find the probability that the team scored (i) exactly 2 goals, (ii) at least 2 goals, (iii) no goal.

(i) 12/40 = **3/10** (1) (ii) 20/40 = **1/2** (1) (iii) 6/40 = **3/20** (1)

28. Find the coordinates of the point on the y-axis which is equidistant from A(−5, 2) and B(3, −2).

Let the point be P(0, y). $PA^2 = 25 + (y-2)^2$, $PB^2 = 9 + (y+2)^2$ (1)
$25 + y^2 - 4y + 4 = 9 + y^2 + 4y + 4 \Rightarrow 29 - 4y = 13 + 4y$ (1) $\Rightarrow y = 2$. Point = **(0, 2)** (1)

29. Simplify: $(a+b)^3 - (a-b)^3$.

$(a+b)^3 = a^3 + 3a^2b + 3ab^2 + b^3$ (1)
$(a-b)^3 = a^3 - 3a^2b + 3ab^2 - b^3$ (1)
Difference = **$6a^2b + 2b^3$** (1)

30. A chord of a circle of radius 15 cm is 18 cm long. Find (i) its distance from the centre; (ii) the length of another chord of this circle at a distance of 9 cm from the centre.

(i) Half chord = 9; d = √(225 − 81) = **12 cm** (1½)
(ii) Half chord = √(225 − 81) = 12 ⇒ chord = **24 cm** (1½)

31. A triangular plot has sides 20 m, 21 m and 29 m. Find its area using Heron's formula.

s = 35 (1); area = √(35 × 15 × 14 × 6) (1) = √44100 = **210 m²** (1)

OR

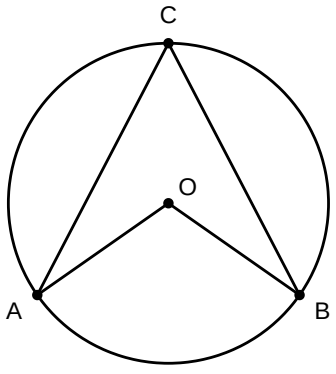
Find the area of a triangle whose sides are 8 cm, 15 cm and 17 cm using Heron's formula.

s = 20 (1); area = √(20 × 12 × 5 × 3) (1) = √3600 = **60 cm²** (1)

SECTION D — Long Answer (5 marks each) — 4 questions · 20 marks

Internal choice is provided in two questions.

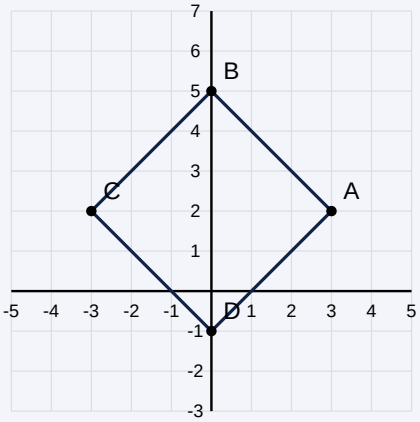
32. Prove that the angle subtended by an arc of a circle at the centre is double the angle subtended by it at any point on the remaining part of the circle (take the case when the arc is a minor arc).



Given: arc AB of a circle with centre O; C a point on the major arc. To prove: ∠AOB = 2∠ACB.
Construction: join CO and produce it to D. (1)

In △OAC, OA = OC ⇒ ∠OAC = ∠OCA; exterior angle ∠AOD = ∠OAC + ∠OCA = 2∠OCA. (1½)
Similarly in △OBC, ∠BOD = 2∠OCB. (1)
Adding: ∠AOD + ∠BOD = 2(∠OCA + ∠OCB) ⇒ ∠AOB = 2∠ACB. (1½)

33. The four corners of a flower bed are marked on a grid as A(3, 2), B(0, 5), C(−3, 2) and D(0, −1).
(a) Plot the points and join them in order. (2 marks)
(b) Find the lengths AB, BC, CD and DA. (1½ marks)
(c) Find the diagonals and justify that ABCD is a square. (1½ marks)



(a) Plotting (1½) + joining (½): (b) AB = √(9 + 9) = 3√2; BC = √(9 + 9) = 3√2; CD = 3√2; DA = 3√2 (1½)
(c) AC = 6, BD = 6 (½). All sides equal and diagonals equal ⇒ ABCD is a square (1)

34. How many multiples of 9 lie between 100 and 500? Find their sum. (2½ + 2½)

Numbers: 108, 117, …, 495 — AP, a = 108, d = 9, l = 495 (1)
n = (495 − 108)/9 + 1 = **44** (1½)
Sum = 44/2 × (108 + 495) = 22 × 603 = **13266** (2½)

O R

The 5th term of an AP is 17 and its 12th term is 45. Find the AP (first three terms) and its 20th term.

a + 4d = 17, a + 11d = 45 (1) ⇒ 7d = 28 ⇒ d = 4 (1) ⇒ a = 1 (1)
AP: **1, 5, 9, …** (1); t_{20} = 1 + 19 × 4 = **77** (1)

35. The breadth of a rectangular field is 7 m less than its length and its area is 120 m². Form a quadratic expression, factorise it to find the length and breadth, and find the perimeter.

Length = x, breadth = x − 7 ⇒ x(x − 7) = 120 ⇒ $x^2 - 7x - 120 = 0$ (1½)
$x^2 - 15x + 8x - 120 = (x - 15)(x + 8) = 0$ (1½) ⇒ x = 15 (1)
Breadth = 8 m; perimeter = 2(15 + 8) = **46 m** (1)

O R

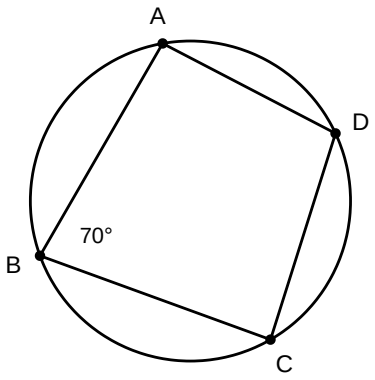
The area of a rectangular carpet is $(6x^2 + 11x + 3)$ m². Factorise the expression to find its length and breadth, and find its perimeter when x = 3.

$6x^2 + 11x + 3 = 6x^2 + 9x + 2x + 3$ (1½) = 3x(2x + 3) + 1(2x + 3) = **(2x + 3)(3x + 1)** (1½)
Dimensions: (3x + 1) m and (2x + 3) m (1). For x = 3: 10 m and 9 m ⇒ perimeter = **38 m** (1)

SECTION E — Case-Study Based Questions (4 marks each) — 3 questions · 12 marks

Read each case carefully and answer the questions that follow. Internal choice is provided in every 2-mark sub-part.

36. A circular running track has four floodlight poles A, B, C and D on its boundary, forming a cyclic quadrilateral ABCD. ∠ABC = 70° and ∠BAD = 85°. O is the centre of the track.



(i) Find ∠ADC.

Opposite angles of a cyclic quadrilateral are supplementary ⇒ ∠ADC = 180° − 70° = **110°**

(ii) Find ∠BCD.

∠BCD = 180° − ∠BAD = 180° − 85° = **95°**

(iii) If arc BC subtends ∠BAC = 30° at A, find ∠BOC.

Angle at the centre is double the angle at the circumference (1) ⇒ ∠BOC = 2 × 30° = **60°** (1)

OR

(iii) A chord of the track is 48 m long and lies 7 m from the centre. Find the radius of the track.

Half chord = 24 m (½); r = √(24² + 7²) (1) = √625 = **25 m** (½)

37. The seats in an auditorium are arranged in rows. The first row has 12 seats and every next row has 2 more seats than the row before it. The number of seats in the n-th row is $t_n = 2n + 10$.

(i) How many seats are there in the 8th row?

$t_8 = 16 + 10 =$ **26**

(ii) Write the common difference of the sequence of seats.

Common difference = **2**

(iii) Which row has 40 seats?

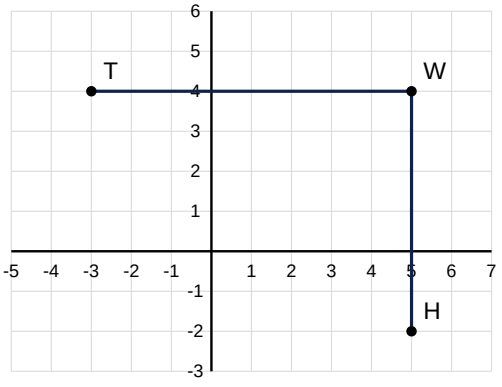
2n + 10 = 40 (1) ⇒ n = **15th row** (1)

OR

(iii) Find the total number of seats in the first 10 rows.

a = 12, l = t_{10} = 30 (1); S = 10/2 × (12 + 30) = **210** (1)

38. The map of a treasure hunt is drawn on a coordinate grid (1 unit = 10 m). The old tree is at T(−3, 4), the well is at W(5, 4) and the hut is at H(5, −2).



(i) Find the distance between the tree and the well (in units).

> TW = $|5 - (-3)|$ = **8 units** (80 m)

(ii) Find the midpoint of the line joining the tree and the hut.

> $\left(\frac{-3+5}{2}, \frac{4-2}{2}\right)$ = **(1, 1)**

(iii) Find the point on the x-axis which is equidistant from the tree and the well.

> P(x, 0): $(x+3)^2 + 16 = (x-5)^2 + 16$ (1) ⇒ $6x + 9 = -10x + 25$ ⇒ $x = 1$; **P(1, 0)** (1)

O R

(iii) Find the straight-line distance between the tree and the hut.

> TH = $\sqrt{[(5+3)^2 + (-2-4)^2]}$ (1) = $\sqrt{100}$ = **10 units** (1)